1 Syntax

$$
\begin{array}{rl}
e ::= & lit \\
& \mid uop\ e \\
& \mid e\ bop\ e \\
& \mid x \\
& \mid e(e_1, \cdots, e_n) \\
& \mid \text{if } e \text{ then } e \text{ else } e \\
& \mid C_i(e_1, \ldots, e_{n_i}) \\
& \mid e \text{ match} \\
& \quad (\text{case } C_i(x_1, \ldots, x_{n_i}) \Rightarrow e)^+ \\
& \mid (x_1 \colon \tau_1, \cdots, x_n \colon \tau_n) \Rightarrow e \\
& \mid (e_1, \cdots, e_n) \\
lit ::= & \text{INTEGER} \mid \text{DOUBLE} \mid \text{STRING} \mid \text{true} \mid \text{false} \mid \text{Nil} \mid prim \\
prim ::= & \text{map} \mid \text{filter} \mid \text{foldLeft} \mid \text{foldRight} \\
uop ::= & - \mid ! \\
bop ::= & + \mid - \mid * \mid / \\
& \mid < \mid > \mid <= \mid >= \mid == \mid != \\
& \mid \&\& \mid || \\
& \mid :: \\
declaration ::= & \text{val } x = e \\
& \mid \text{type } T = \tau \\
& \mid \text{def } f(x_1 \colon \tau_1, \cdots, x_n \colon \tau_n) \colon \tau = e \\
& \mid \text{def } f[\alpha_1, \cdots, \alpha_m](x_1 \colon \tau_1, \cdots, x_n \colon \tau_n) \colon \tau = e \\
& \mid \text{enum } T[\alpha_1, \cdots, \alpha_m] \colon \\
& \quad (\text{case } C_i(x_1 \colon \tau_1, \ldots, x_{n_i} \colon \tau_{n_i}))^+ \\
\tau ::= & \text{int} \mid \text{double} \mid \text{String} \mid \text{Bool} \mid \text{List}[\tau] \mid \tau \to \tau \mid (\tau_1, \cdots, \tau_n) \mid T \mid \alpha
\end{array}
$$

2 Type Checking

Value Reference

$$\frac{x : \tau \in \Gamma}{\Gamma \vdash x : \tau}\ \text{T-ValueRef}$$

Arithmetic

$$\frac{}{\Gamma \vdash n : \mathit{int}}\ \text{T-Int} \qquad \frac{}{\Gamma \vdash n : \mathit{double}}\ \text{T-Double}$$

$$\frac{\Gamma \vdash e : \tau \qquad \tau \in \{\text{int}, \text{double}\}}{\Gamma \vdash -e : \tau}\ \text{T-ArithNeg}$$

$$\frac{\Gamma \vdash e_1 : \tau \qquad \Gamma \vdash e_2 : \tau \qquad \tau \in \{\text{int}, \text{double}\}}{\Gamma \vdash e_1 + e_2 : \tau}\ \text{T-Add}$$

$$\frac{\Gamma \vdash e_1 : \tau \qquad \Gamma \vdash e_2 : \tau \qquad \tau \in \{\text{int}, \text{double}\}}{\Gamma \vdash e_1 - e_2 : \tau}\ \text{T-Sub}$$

$$\frac{\Gamma \vdash e_1 : \tau \qquad \Gamma \vdash e_2 : \tau \qquad \tau \in \{\text{int}, \text{double}\}}{\Gamma \vdash e_1 * e_2 : \tau}\ \text{T-Mul}$$

$$\frac{\Gamma \vdash e_1 : \tau \qquad \Gamma \vdash e_2 : \tau \qquad \tau \in \{\text{int}, \text{double}\}}{\Gamma \vdash e_1 / e_2 : \tau}\ \text{T-Div}$$

$$\frac{\Gamma \vdash e_1 : \tau \qquad \Gamma \vdash e_2 : \tau \qquad \tau \in \{\text{int}, \text{double}\}}{\Gamma \vdash e_1 < e_2 : \tau}\ \text{T-LT}$$

$$\frac{\Gamma \vdash e_1 : \tau \qquad \Gamma \vdash e_2 : \tau \qquad \tau \in \{\text{int}, \text{double}\}}{\Gamma \vdash e_1 > e_2 : \tau}\ \text{T-GT}$$

$$\frac{\Gamma \vdash e_1 : \tau \qquad \Gamma \vdash e_2 : \tau \qquad \tau \in \{\text{int}, \text{double}\}}{\Gamma \vdash e_1 <= e_2 : \tau}\ \text{T-LTEQ}$$

$$\frac{\Gamma \vdash e_1 : \tau \qquad \Gamma \vdash e_2 : \tau \qquad \tau \in \{\text{int}, \text{double}\}}{\Gamma \vdash e_1 >= e_2 : \tau}\ \text{T-GTEQ}$$

Strings

$$\frac{}{\Gamma \vdash s : \mathit{string}}\ \text{T-String}$$

Boolean Logic

$$\frac{}{\Gamma \vdash \mathit{true} : \mathit{bool}}\ \text{T-True} \qquad \frac{}{\Gamma \vdash \mathit{false} : \mathit{bool}}\ \text{T-False} \qquad \frac{\Gamma \vdash e : \mathit{bool}}{\Gamma \vdash !e : \mathit{bool}}\ \text{T-BoolNeg}$$

$$\frac{\Gamma \vdash e_1 : \mathit{bool} \qquad \Gamma \vdash e_2 : \mathit{bool}}{\Gamma \vdash e_1 \&\& e_2 : \mathit{bool}}\ \text{T-And} \qquad \frac{\Gamma \vdash e_1 : \mathit{bool} \qquad \Gamma \vdash e_2 : \mathit{bool}}{\Gamma \vdash e_1 || e_2 : \mathit{bool}}\ \text{T-Or}$$

Equality Comparison

$$\frac{\Gamma \vdash e_1 : \tau \qquad \Gamma \vdash e_2 : \tau}{\Gamma \vdash e_1 == e_2 : \mathit{bool}}\ \text{T-EQ} \qquad \frac{\Gamma \vdash e_1 : \tau \qquad \Gamma \vdash e_2 : \tau}{\Gamma \vdash e_1 != e_2 : \mathit{bool}}\ \text{T-NEQ}$$

Tuples

$$\frac{\Gamma \vdash e_i : \tau_i \quad (1 \le i \le n)}{\Gamma \vdash (e_1, \ldots, e_n) : (\tau_1, \ldots, \tau_n)}\ \text{T-Tuple}$$

Lists

$$\frac{}{\Gamma \vdash \mathit{Nil} : \tau : \mathit{List}[\tau]}\ \text{T-Nil} \qquad \frac{\Gamma \vdash e_1 : \tau \qquad \Gamma \vdash e_2 : \mathit{list}[\tau]}{\Gamma \vdash e_1 :: e_2 : \mathit{list}[\tau]}\ \text{T-Cons}$$

Control Flow

$$\frac{\Gamma \vdash e : \mathit{bool} \qquad \Gamma \vdash e_1 : \tau \qquad \Gamma \vdash e_2 : \tau}{\Gamma \vdash \mathit{if}\ e\ \mathit{then}\ e_1\ \mathit{else}\ e_2 : \tau}\ \text{T-If}$$

Functions

$$\frac{\Gamma, x_1 : \tau_1, \ldots, x_n : \tau_n \vdash e : \tau}{\Gamma \vdash (x_1 : \tau_1, \ldots, x_n : \tau_n) \Rightarrow e\ :\ (\tau_1, \ldots, \tau_n) \to \tau}\ \text{T-Lambda}$$

$$\frac{\Gamma \vdash e : (\tau_1, \ldots, \tau_n) \to \tau \qquad \Gamma \vdash e_i : \tau_i \quad (1 \le i \le n)}{\Gamma \vdash e(e_1, \ldots, e_n) : \tau}\ \text{T-Apply}$$

Primitives

$$\frac{}{\Gamma \vdash \text{map} : \forall A, B.\ (A \to B) \to \text{List}[A] \to \text{List}[B]}\ \text{T-Map}$$

$$\frac{}{\Gamma \vdash \text{filter} : \forall A, B.\ \text{List}[A] \to (A \to \text{Bool}) \to \text{List}[A]}\ \text{T-Filter}$$

$$\frac{}{\Gamma \vdash \text{foldLeft} : \forall A, B.\ \text{List}[A] \to B \to ((B, A) \to B) \to B}\ \text{T-FoldLeft}$$

$$\frac{}{\Gamma \vdash \text{foldRight} : \forall A, B.\ \text{List}[A] \to B \to ((A, B) \to B) \to B}\ \text{T-FoldRight}$$

Constructor

$$\frac{\Gamma \vdash C_i : (\tau_1, \ldots, \tau_{n_i}) \to T[\alpha_1, \ldots, \alpha_m] \qquad \Gamma \vdash e_j : \tau_j[\bar{\alpha} := \bar{\sigma}] \quad (1 \leq j \leq n_i)}{\Gamma \vdash C_i(e_1, \ldots, e_{n_i}) : T[\sigma_1, \ldots, \sigma_m]} \text{ T-ConstrApply}$$

Match

$$\frac{\begin{array}{c}\Gamma \vdash e : T[\sigma_1, \ldots, \sigma_m] \\ \big(\Gamma \vdash C_i : (\tau_{i,1}, \ldots, \tau_{i,n_i}) \to T[\alpha_1, \ldots, \alpha_m]\big)^+ \\ \big(\Gamma, x_1 : \tau_{i,1}[\bar{\alpha} := \bar{\sigma}], \ldots, x_{n_i} : \tau_{i,n_i}[\bar{\alpha} := \bar{\sigma}] \vdash e_i : \tau\big)^+ \\ \{C_1, \ldots\} = \text{constructors}(T)\end{array}}{\Gamma \vdash e \text{ match } (\text{case } C_i(x_1, \ldots, x_{n_i}) \Rightarrow e_i)^+ : \tau} \text{ T-Match}$$

Declarations

$$\frac{\Gamma \vdash e : \tau}{\Gamma \vdash \text{val } x = e \dashv \Gamma, x : \tau} \text{ T-ValDecl} \qquad \frac{}{\Gamma \vdash \text{type } T = \tau \dashv \Gamma, T = \tau} \text{ T-TypeDecl}$$

$$\frac{\Gamma, x_1 : \tau_1, \ldots, x_n : \tau_n \vdash e : \tau}{\Gamma \vdash \text{def } f(x_1 \colon \tau_1, \ldots, x_n \colon \tau_n) \colon \tau = e \dashv \Gamma, f : (\tau_1, \ldots, \tau_n) \to \tau} \text{ T-DefDecl}$$

$$\frac{\Gamma, \alpha_1, \ldots, \alpha_m, x_1 : \tau_1, \ldots, x_n : \tau_n \vdash e : \tau}{\Gamma \vdash \text{def } f[\alpha_1, \ldots, \alpha_m](x_1 \colon \tau_1, \ldots, x_n \colon \tau_n) \colon \tau = e \dashv \Gamma, f : \forall \alpha_1, \ldots, \alpha_m.\ (\tau_1, \ldots, \tau_n) \to \tau} \text{ T-DefDeclPoly}$$

$$\frac{\begin{array}{c}\Gamma' = \Gamma, \alpha_1, \ldots, \alpha_m \\ \Gamma'' = \Gamma', T[\alpha_1, \ldots, \alpha_m] \\ \big(\Gamma'' \vdash C_i : (\tau_{i,1}, \ldots, \tau_{i,n_i}) \to T[\alpha_1, \ldots, \alpha_m]\big)^+\end{array}}{\begin{array}{l}\Gamma \vdash \text{enum } T[\alpha_1, \ldots, \alpha_m] \colon \ (\text{case } C_i(x_1 \colon \tau_1, \ldots, x_{n_i} \colon \tau_{n_i}))^+ \dashv \\ \Gamma, T[\alpha_1, \ldots, \alpha_m], \big(C_i : (\tau_{i,1}, \ldots, \tau_{i,n_i}) \to T[\alpha_1, \ldots, \alpha_m]\big)^+\end{array}} \text{ T-EnumDecl}$$